Effects of Video Modeling and Verbal Prompting on Social Skills Embedded Within a Purchasing Activity for Students with Autism

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Three junior high school students with autism were taught to engage in a series of social skills through the use of video modeling and verbal prompting as they engaged in purchasing activities in two grocery stores. A multiple baseline across students design was used to demonstrate the effectiveness of video modeling with verbal prompting on social skill acquisition within a purchasing task. Students watched the video modeling clip once during each intervention session for three weeks and demonstrated increased use of target skills in a grocery store. For generalization, skills were assessed in a second grocery store over three sessions for two weeks. Results indicated one student was successful in acquiring and generalizing social skills to a second untrained setting while two students demonstrated variable results.

Individuals who successfully develop social skills demonstrate greater independence, social acceptance, adaptive behavior, and employment success (Banda & Hart, 2010; Scott, Clark, & Brady, 2000). When social skill discrepancies exist, they may lead to failures in employment and community functioning (Fredericks, Buckley, Baldwin, Moore, & Stremel-Campbell, 1983). For individuals with autism spectrum disorders (ASD), deficits in social interactions are significant; they include difficulties in social initiation and in developing and maintaining social relationships (American Psychiatric Association [APA], 2000). Individuals with ASD may struggle with engaging in conversations, developing friendships, and participating in cooperative play with peers (Scott et al.). While social skills research and instruction is an area of need for those with ASD (Howlin, 2005; Shea & Mesibov, 2005), it remains a less frequently studied area than others in the field (Reichow & Volkmar, 2010).

Social skill development for individuals with ASD and/or developmental disabilities was the focus of investigations in which social stories (Thiemann & Goldstein, 2001), pivotal response training (Pierce & Schreibman, 1997), live peer models in real life settings (Hundert & Houghton, 1992), incidental teaching (McGee & Daly, 2007), and picture exchange communication system (Charlop-Christy, Carpenter, Le, LeBlanc, & Kellet, 2002) were used. Each strategy proved to be effective in increasing a range of skills such as social-communication (Charlop-Christy et al.; Sansosti & Powell-Smith, 2008), maintaining interactions, initiating conversations, and initiating play (Pierce & Schreibman; Thiemann & Goldstein).

Whereas numerous strategies demonstrate potential for increasing appropriate social skills and behaviors with individuals with ASD (e.g., backward chaining, videotaping, peer modeling, pivotal response training, telling social stories), interventions infusing technology also are being used. Video technology is used increasingly for facilitating increases in a wide range of skills—such as daily living and social and academic skills—for individuals with ASD (McCoy & Hermansen, 2007). Video simulation strategies combining audio and video technologies specifically targeting social skill acquisition for these individuals were found to be effective when presented as video models and prompts (Deitchman, Reeve, Reeve, & Progar, 2010; Maione & Mirenda, 2006). Video modeling requires individuals to watch an entire task chain presented in a video clip prior to task performance (Cannella-Malone, Sigafoos, O'Reilly, de la Cruz, Edrisinha, & Lancioni, 2006).

Individuals with ASD used video modeling effectively to increase skills in grocery store settings (Alcantara, 1994; Ayres & Langone, 2007), perspective taking skills (Charlop-Christy & Daneshvar, 2003), functional skills (Shipley-Benamou, Lutzker, & Taubman, 2002), and vocational skills (Cihak & Schrader, 2008). When teaching social skills (Buggey, 2005; Wert & Neisworth, 2003), social initiations (Maione & Mirenda, 2006; Nikopoulos & Keenan, 2004) and conversational skills (Buggey, Toombs, Gardener, & Cervetti, 1999; Sherer, Pierce, Paredes, Kisacky, Ingersoll, & Schreibman, 2001), video modeling was used mainly with elementary school children in play settings and/or school environments. For improving social skills, Nikopolous and Keenan (2003) found video modeling to be efficient and effective in increasing and generalizing social behaviors when examined across multiple conditions.

Social skills and other social behaviors used within the context of community functioning lead to increased independence (Haring, Kennedy, Adams, & Pitts-Conway, 1987; Westling, Floyd, & Carr, 1990). Daily living skills frequently necessitate effective use of communication skills when ordering in a fast food restaurant (Mechling, Pridgen, & Cronin, 2005) and purchasing items in convenience and grocery stores (Ayres & Langone, 2002; Wissick, Lloyd, & Kinzie, 1992). Within a grocery setting, these skills include standing in a checkout line without interfering with other customers; responding to cashier requests when paying for a purchase; identifying a preferred bag (e.g., paper or plastic); and retrieving change, receipts, and groceries (Hansen & Morgan, 2008).

Previous research demonstrated that skills and specific social behaviors could be taught successfully as part of a purchasing sequence. For example, Mechling (2004) taught three students with a mild to moderate intellectual disability reading fluency while concurrently locating grocery items; Cihak and Grim (2008) taught students with autism and moderate intellectual disability money skills; McDonnell and Laughlin (1989) taught students with a moderate and/or severe intellectual disability to order, purchase, and eat items in fast food restaurants; and investigators taught students with autism a sequence of purchasing activities from entering to exiting the store (Alcantara, 1994; Haring, Breen, Weiner, Kennedy, & Bednersh, 1995). In addition to teaching traditional skills used in a purchasing activity, including other related skills in purchasing skills instruction (e.g., teaching nutrition, making a shopping list, correct use of money to pay for items, asking for a specific item, social skills, interacting with the cashier, turn taking, and waiting in line) provides students with a broad range of opportunities to practice and acquire skills necessary to function independently in community settings (Morse, Schuster, & Sandknop, 1996).

Examples of specific behaviors and skills that result in social competence and increase a child's effectiveness in participating in daily life include demonstrating good manners by using terms such as "please" and "thank you," smiling, using and initiating greetings, taking turns in conversation with others, and demonstrating appropriate behaviors during mealtime (Hurley, Wehby, & Feurer, 2010). Studies focusing on embedding social responses and behaviors within purchasing skills demonstrated successful use of video-based instruction in teaching appropriate greeting, social response to the cashier, and saying "thank you" along with purchasing (Haring et al., 1987), appropriate social behaviors (Westling et al., 1990), saying "thank you" during purchasing skills instruction (Alcantara, 1994), and in ordering food (Mechling & Cronin, 2006).

Providing instruction in community-based settings is considered an instructional best practice for students with disabilities (Alberto, Cihak, & Gama, 2005). Community-based instruction provides students with disabilities an opportunity to acquire skills in the settings in which they will be used, thus increasing the likelihood for skill generalization (Cihak, Alberto, Kessler, & Taber, 2004). When used concurrently, video modeling and in-vivo instruction was suggested to be effective in achieving more independence in completing functional skills (e.g., putting on a wrist watch, mailing a letter, crossing the street; Branham, Collins, Schuster, & Kleinert, 1999); locating books and DVDs in a public library (Taber-Doughty, Patton, & Brennan, 2008); and developing community skills such as grocery purchasing and using automated teller machines to withdraw money (Alberto et al., 2005; Cihak, Alberto, Taber-Doughty, & Gama, 2006; Mechling, Gast, & Barthold, 2003). However, there are few studies that focus on embedding social behaviors and skills within a variety of community functioning skills to individuals with autism and intellectual disability in community settings (e.g., Cihak, Alberto, & Fredrick, 2007; Westling et al., 1990).

Training students in a skill such as making purchases within community settings should be considered a priority, as it not only increases access and participation in new environments (Ayres & Langone, 2002; Hansen & Morgan, 2008), but it also promotes autonomous behavior in those environments (Hutcherson, Langone, Ayres, & Clees, 2004). When combined with skills such as social greetings, maintaining eye contact, and other socially appropriate behaviors, the purchasing experience may be more greatly enhanced, leading to increases in positive human interactions.

The purpose of this study was to examine the effectiveness of video modeling combined with verbal prompting on social skills within a purchasing task sequence with three junior high school students with autism when making purchases in local grocery stores. Specifically, students were taught to stand in line, greet a cashier, smile, pay, say "thank you," and retrieve the purchased items following video modeling instruction. Following intervention, skills were then assessed for generalization in a second grocery setting.

Method

Participants

Three students, Tom, Steve, and Nick participated in this study. Each was selected to participate based on the following criteria: (a) a primary diagnosis of autism; (b) a junior high school student age 12–15; (c) the need to improve purchasing and social skills in community settings; (d) no prior experience with video modeling instruction; and (e) willingness to participate. Each student attended a suburban junior high school in a Midwestern community; participated in community-based instruction two days per week, one hour per day; and received daily instruction in a self-contained life skills classroom. During regular classroom trips to a grocery store as part of community-based instruction, students used a self-checkout lane to purchase items via debit/credit card. The instruction students received involved verbal instruction and physical guidance from the classroom teacher. Students did not receive instruction using video modeling to purchase items through cashier's checkout lane and paying with cash prior to and during the study. Table 1 provides a summary for each student.

Tom was a 14-year-old male diagnosed with mild to moderate autism (Gilliam Autism Rating Scale [GARS-2]) as a primary diagnosis and a moderate intellectual disability (Wechsler Intelligence Scale for Children IV [WISC-IV] IQ 46) as a secondary diagnosis. He was able to communicate his needs by composing full sentences. He was eager to learn and took time to concentrate on each task. When given directions he could follow written, verbal, and visual directions well. He was friendly and enjoyed being around his peers. While he engaged in frequent verbal communication with familiar people, he avoided communicating with unfamiliar individuals in community

Table 1

Participant Characteristics

Name	Gender	Age	Autism Diagnosis	IQ Scores	Adaptive Behavior Assessment	Community-based Instruction
Tom	Male	14	Mild/moderate autism; GARS-2	IQ-46 (WISC-IV)	ABI-48* (ABAS-II†)	2 hours per week
Steve	Male	15	Autism/intermittent explosive disorder; DSM-4	IQ-40 (KBIT-II)	ABI-43 (ABAS-II)	2 hours per week
Nick	Male	12	Mild/moderate autism; CARS	IQ-57 (UNIT)	ABI- 40 (ABAS-II)	2 hours per week

*** ABI: Adaptive Behavior Inventory**

† ABAS-II: Adaptive Behavior Assessment System-II

settings by keeping his fingers in his mouth and avoiding eye contact. He frequently invaded the personal space of others by standing very close to them when talking. When purchasing items in a grocery store he was able to locate items independently, find the self-checkout lane, and use the self-checkout lane with the teacher's assistance. He did not experience interactions with cashiers.

Steve was a 15-year-old male with a primary diagnosis of autism, and a secondary diagnosis of intermittent explosive disorder (Diagnostic and Statistical Manual of Mental Disorders [DSM] -IV) and a moderate intellectual disability (Kaufman Brief Intelligence Test -11 [KBIT-II] IQ 40) who spoke only a few words, such as "hi," "ok," and "yes." He primarily communicated his needs and feelings using American Sign Language. Steve did not recognize the physical space of others and did not make eye contact. He did not interact with peers appropriately or initiate or respond to peers' requests, and he had little to no interaction with them. He also did not interact with staff and other familiar adults in the school. Steve frequently repeated his preferred words, such as "dad," "bed," and "food," and when upset or anxious he would cry, pull on his clothes, flap his hands rapidly, and look around the physical space without visually attending to the task at hand. He took frequent walks or squeezed toys to calm down. However, when focused, he could follow verbal and visual directions well. While in a grocery store he was able to locate items with a familiar location independently, but he needed assistance in locating other items. He also was able to locate the self-checkout lane independently and use the self-checkout lane with the teacher's assistance. He did not interact with others in a community setting.

Nick was a 12-year-old male with a primary diagnosis of mild to moderate autism (Childhood Autism Rating Scale [CARS]) and a secondary diagnosis of a moderate intellectual disability (Universal Nonverbal Intelligence Test [UNIT] IQ 57). He was nonverbal and used a Dynavox V speech-generating device to speak familiar phrases, but he occasionally generated new utterances. He often sat at his desk and rubbed his nose or he got up and paced. He went for walks throughout the day to help relieve any stress or anxiety. He also had difficulty with distractions. However, when he was focused he could follow verbal, written, and visual directions well. Nick's communication was considerably restricted due to his lack of spontaneous speech, even with his speech-generating device. He avoided interactions with both familiar and unfamiliar people, and rarely engaged in communication with classroom teachers and peers. He frequently engaged in hand flapping, and he avoided physical and eye contact when interacting with others. He had previous experience using the self-checkout lane at a grocery store, but did not use the cashier's checkout lane. He was able to locate items and find the self-checkout lane independently and use the self-checkout lane with the teacher's assistance. He experienced limited interaction with unfamiliar people in a grocery store setting.

Settings

Activities took place in three settings: the self-contained classroom and two local grocery stores. The classroom was divided into two sections, with two desktop computers located on a table on one side of the room. On the other side were tables and student work stations. An additional computer was located close to the teacher's desk. This was where students viewed video clips. Nine students, four paraprofessionals, a nurse, and a licensed special education teacher were present in the room on most school days. In the community, baseline and intervention sessions occurred in the checkout line of a large chain superstore, while generalization sessions took place in a second similar yet novel local grocery store. In each store there were up to eight checkout lines; three-four had cashiers, and two-four were self-checkout lines. During each grocery visit, students generally entered the first or second open lane with a cashier. There were approximately one or two customers in line before and after students during each purchasing activity throughout baseline, intervention, and generalization sessions.

Independent Variable

Video modeling and verbal prompts were used as independent variables in the study. Video modeling was defined as watching a video of an individual performing a series of target tasks and then performing those tasks in the actual setting. Verbal prompts involved providing the student with verbal reminders if he missed or performed any task step incorrectly during the purchasing activity. The video illustrated an adult approaching the checkout lane and proceeding through a series of steps while emphasizing the target social and purchasing skills. The video was a minute and 10 seconds in length. It included an audio recording of the adult saying "stay in line" and "wait for the turn to come." The video also included the

conversation between the cashier and the model during the greeting, payment transaction, and thanking exchanges. Each student viewed the video.

Dependent Variables

Standing in line, greeting the cashier, smiling, paying the cashier, thanking the cashier, and retrieving purchased grocery items at the end of the transaction were the dependent variables. Students were scored as independent when they met the following criteria for each dependent variable:

1. **Standing in line**. The student located and stood in the checkout lane and waited for his turn to check out. He remained in line, did not move in front of other customers, and moved forward to checkout when his turn approached.
2. **Greeting the cashier**. The student greeted the cashier by responding to the cashier's greeting using his method of communication. Several responses were considered an acceptable greeting—for example, if the student said, "hi," "hello," or "good morning" in response to the cashier's greeting. In addition, if the student answered "good" to the cashier's initiation of "how are you?" this was recorded as an appropriate greeting.
3. **Smiling**. This was recorded if the student physically smiled (e.g., demonstrated an upward curling of the mouth) when he approached and greeted the cashier as well as during the final check out.
4. **Paying the cashier**. The student paid for the items with cash by handing money to the cashier and receiving any change.
5. **Thanking the cashier**. At the end of the transaction, the student said "thank you" to the cashier, or said "thank you" in response to the cashier's thanking him.
6. **Retrieving the purchased grocery items at the end of the transaction**. The student got the items at the end of the transaction.

Experimental Design

A multiple baseline across students design (Kazdin, 1982) was used to assess the effectiveness of video modeling and verbal prompting on increases in social and purchasing skills of students with ASD as they participated in a purchasing activity in a grocery store. This design was selected because it enabled investigators to establish a functional relationship between the independent and dependent variables as the pattern of change was replicated across students between baseline and intervention conditions (Kazdin; Kennedy, 2005). Baseline sessions consisted of a minimum of five sessions, and treatment began with one student to determine the effect of the independent variable while the other two students continued with baseline. Replication of effect with each student was established by progressively introducing intervention to each student. A functional relationship was demonstrated when an increase in the use of target skills upon introduction of intervention was observed. In addition, generalization to a second setting confirmed that students had generalized the skills after training in one setting. This design permitted researchers to evaluate continued demonstration of intervention effects and generalization of skills into untrained settings.

Data Collection

Event recording was used to determine the percentage of social and purchasing skills demonstrated independently by each student while engaging in shopping checkout behaviors. This measurement technique was selected because it allowed investigators to easily record behaviors with brief durations and provided a precise estimate of behaviors to evaluate the desired response (Kennedy, 2005). Behavior was recorded as occurrence or nonoccurrence. Occurrences were recorded as correct independent performance of a task step with no prompts provided by the experimenter. Nonoccurrences were recorded when a student missed a step of the task, performed incorrectly, or required additional prompting.

Procedures

During baseline, intervention, and generalization sessions investigators delivered directions, the video modeling instruction in the classroom, and verbal prompts when necessary. The purpose of providing verbal prompts was to allow students the opportunity to complete the task in the community setting without having to watch the video modeling clip again. Verbal prompts were provided only when a student performed a step incorrectly or was unresponsive during the purchasing sequence. However, students did not require verbal prompts on all task steps. During all phases, after students located the checkout lane, the investigator stood at the bagging side of the checkout lane about one meter away from the student. She used a

data sheet with a task analysis to record the occurrences and nonoccurrences of each task step.

Baseline. This phase took place at a local grocery store during regularly scheduled community-based instruction and occurred until stable responding was achieved. Students used a public bus system to travel to a local grocery store from their school. During each baseline session, after students retrieved items from their shopping list, the investigator instructed students to proceed to the cashier's checkout lane to purchase items. Students did not receive intervention or additional instructions during baseline sessions. During the checkout routine, each student's target behaviors were recorded as occurrence or nonoccurrence for standing in line, greeting the cashier, smiling, following cashier directions, paying, thanking the cashier, and retrieving the purchased grocery items after paying. Baseline data were collected for five sessions for Tom, eight sessions for Steve, and 10 sessions for Nick. No prompts were given if a student missed a step of the task sequence, and only independent and correct performance of task steps was recorded as occurrence.

Intervention. Intervention sessions occurred two days a week for three weeks and lasted six sessions per student. Each intervention session involved each student viewing the video recording of an adult modeling the social and purchasing skills when engaged in the grocery checkout routine. Prior to watching the video during the first intervention session, each student was instructed to watch the entire video and pay attention to the model and the task. Each student also was told that he needed to perform the same activities the person in the video performed. During subsequent intervention sessions, students were told that they would be watching the same entire video clip once, as in the previous sessions, and were given the same directions as in the first intervention session. The investigator did not direct students' attention to any specific aspects of the video clip and did not provide any instructions as to the specific skills with which students were struggling.

Approximately one hour prior to engaging in the purchasing activity, students viewed the entire video clip on the classroom computer. The one-hour time interval was chosen to analyze its effects on video instruction. While in the grocery store, each student located the items on a shopping list. Then the investigator instructed the students to proceed to the cashier's checkout lane to purchase the items. Each student was told to act and do the same activities according to the video modeling clip. Data were collected on the target behaviors for each student. Steve and Nick used their preferred method of communication during purchasing task, and independent performance of task steps using their communication modes was marked as correct. During this condition, verbal prompts were available to each student if he stopped in the middle of the task or missed or performed any step of the task incorrectly. In such instances, the prompt, "What do you need to do now?" was used. If the student did not respond within a few seconds, he was provided a verbal model. For example, a verbal model for a missed or incorrect greeting task was asking, "Do you remember you should say hello when you approach the cashier?" Any step performed with a verbal prompt was considered incorrect and the behavior was coded as a nonoccurrence.

Generalization. To assess the generalization of learned skills, students performed the purchasing routine with embedded social skills in a second untrained grocery store for three sessions. As in previous phases, students used a public bus system to travel to a grocery store from the school. After each student located the items on his shopping list, the investigator instructed students to proceed to the cashier's checkout lane to purchase items. Students completed the same purchasing tasks without watching the video clip prior to engaging in the grocery checkout routine. In addition, no verbal prompts were available during generalization sessions. Nick began this condition immediately following intervention. Steve began one week following intervention, and Tom began two weeks following intervention. Independent performance of task steps was recorded as occurrence, and any task step not performed or performed incorrectly was marked as nonoccurrence.

Inter-observer Agreement

To ensure data validity, two independent observers documented the dependent variables independently during 30% of each condition. Agreement was calculated by dividing the number of recorded agreements by the number of agreements plus disagreements and then multiplying by 100%. Inter-observer agreement was 100% for Tom and Steve across all conditions. Nick's mean inter-observer agreement score was 94.4% during baseline with a range of 83.3%–100%; 92% during intervention with a range of 91.6%–100%; and 100% during generalization phase.

Treatment Fidelity

Treatment fidelity measures confirmed the correct use of intervention procedures. Using a task analysis, a trained classroom paraprofessional familiar with the intervention steps observed the number of steps implemented correctly. Objectives and procedures as well as the intervention were explained to the paraprofessional prior to the beginning of the study. The paraprofessional verified each step, from the video modeling intervention through the implementation of the purchasing task in community settings. When examining consistency with which video modeling was implemented, the paraprofessional determined whether (1) each student was told that he would be watching a video clip of an adult purchasing in a grocery store; (2) each student watched the video clip prior to his arrival at the grocery store; (3) each student watched the video clip only a single time during each intervention session; and (4) each student was instructed to complete the same activities he viewed in the video clip. Treatment fidelity was calculated by dividing the number of completed steps (occurrences) by the number of completed and uncompleted steps (nonoccurrences) and then multiplying by 100%. Treatment fidelity was 100% for each student during intervention sessions.

Social Validity

An informal interview with students was conducted following completion of study activities to determine the social validity of the intervention and target behaviors. Students were asked about their likes and dislikes of using video modeling for completing the purchasing routine. Students responded to the questions with "yes/no" answers using their method of communication. Table 2 presents a list of questions students answered.

Additionally, the classroom teacher completed a satisfaction assessment using a 5-point Likert-like scale to determine her satisfaction with using video modeling as an intervention for teaching social along with grocery shopping skills to her students with ASD. She was asked to identify how satisfied she felt that students enjoyed watching the video clip, the intervention was easy to implement, students easily performed the tasks in a grocery store, and the effects of the intervention were positive on students' skill acquisition. She also indicated how likely she was to consider using video modeling in her class in the future to teach social behaviors. Finally, the grocery store cashiers were interviewed briefly to obtain their opinions on each student's performance during the checkout routine.

Table 2

Student Social Validity Questions

Post intervention questions
1. Did you like watching the video clip?
2. Was the video clip easy to understand?
3. Did you feel comfortable in the grocery store when you performed the tasks?
4. Do you think you could use these skills from the video in the community (i.e. dining places, malls, on public transportation across the community settings)?

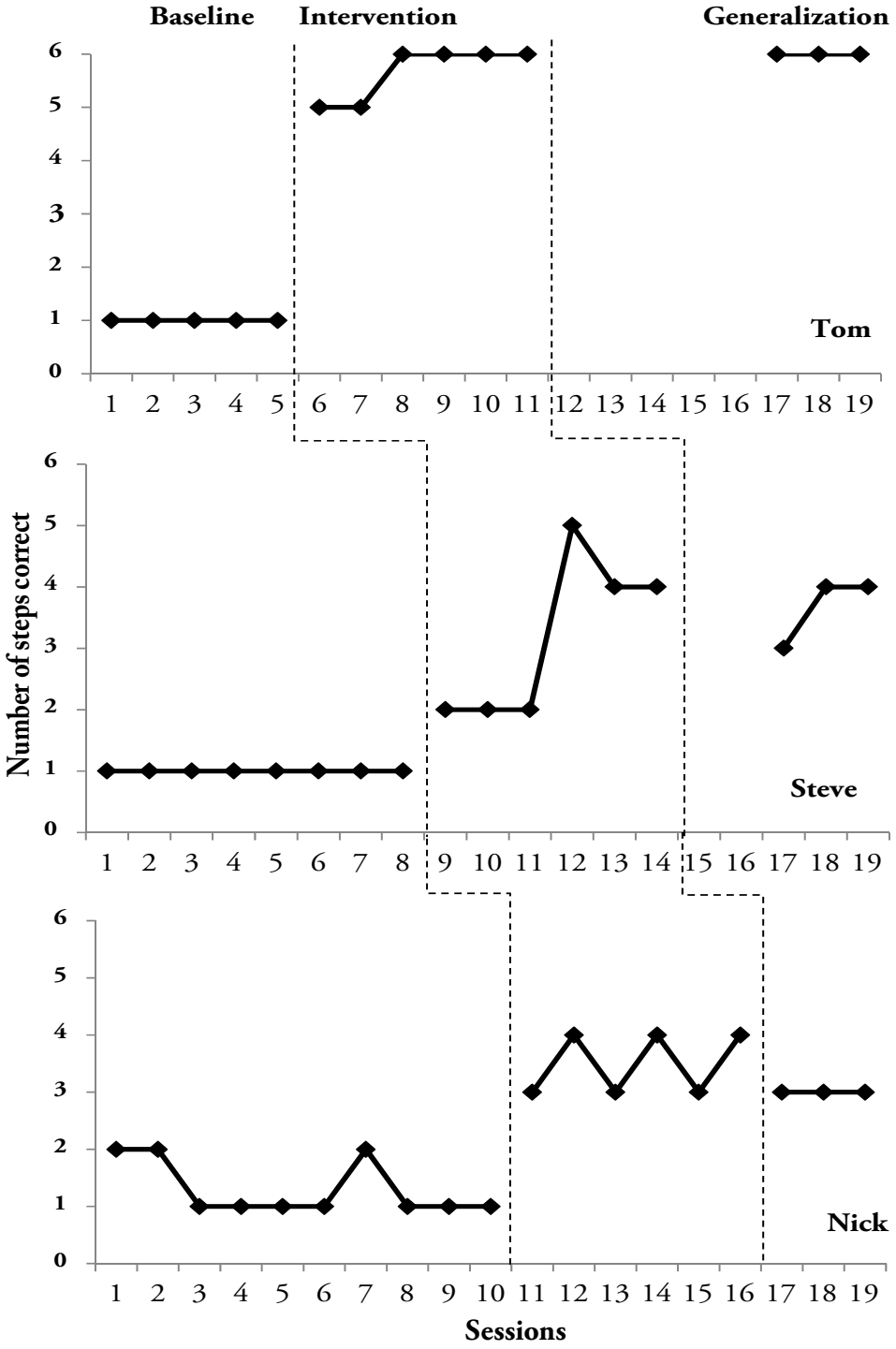
Results

Figure 1 illustrates the effectiveness of video modeling and verbal prompting on independent performance of social and purchasing skills of three junior high school students with autism when making purchases in two community settings. The figure reflects only the independent correct performance of each task step (occurrences). Visual analysis confirmed the existence of a functional relationship between the independent and dependent variables with the replication of effect demonstrated for each student. Analysis also indicated the intervention resulted in increases in each student's performance between baseline and intervention conditions. When generalizing purchasing and social skills to an untrained grocery store setting, the mean level of independent performance increased over baseline levels for all three students.

Tom. During baseline, Tom demonstrated a low and stable level of performance. He achieved a baseline mean of 1.0 when engaging in social and shopping skills during purchasing activities. During baseline phase, he performed one out of six steps correctly—paying the cashier. With the introduction of intervention, a rapid increase in Tom's performance was observed, with 100% of percentage of nonoverlapping data (PND) between baseline and intervention sessions. The mean change between baseline and intervention phases was 4.67. He completed five of six steps correctly during the first two intervention sessions, and his performance increased to 100% during the last four

Figure 1

Number of steps completed independently during a purchasing task.



sessions. During the first intervention session he missed "smiling," and during the second intervention session he did not perform "greeting the cashier" independently. Thus, his mean level of performance during intervention was 5.67 when video modeling was used. During generalization, Tom achieved a mean performance level of 6.0 and 100% independence during three consecutive sessions. This was slightly higher, demonstrating a mean change of 0.33% above that achieved during intervention.

Steve. Steve's baseline performance remained stable, with a mean performance score of 1.0 achieved throughout this condition in displaying social and purchasing skills while grocery shopping. During this phase, Steve performed one out of six steps correctly, i.e. "paying the cashier." With the introduction of video modeling his intervention mean increased to 3.17, indicating a mean change in performance level of 2.17 between baseline and intervention sessions. Steve's performance between baseline and intervention phases indicated a PND of 100%. A moderate degree of variability in data points was observed during the intervention session. During the first two intervention sessions, he independently performed "standing in line" and "paying the cashier." Steve could say "hi," or "hello," but physically signed when he thanked the cashier. He independently performed "greeting the cashier" during one intervention session, and independently thanked the cashier during three intervention sessions. For retrieving purchased items, he required verbal prompting during two sessions of the intervention phase. Steve did not demonstrate "smiling" throughout the entire intervention phase. He demonstrated a slowly increasing change in the performance of skills for three consecutive sessions, and then a rapid increase in the rate of performance. During generalization, he demonstrated a slight increasing trend and stable level of performance with a mean performance of 3.67. During the first session of generalization he independently performed "standing in line," "paying the cashier," and "retrieving purchased items." During the last two generalization sessions he independently performed four steps of the task correctly— "standing in line," "greeting the cashier," "paying," and "retrieving purchased items."

Nick. Nick achieved a baseline mean of 1.3 when initially engaged in social and purchasing skills during the grocery shopping routine. During all sessions of baseline, he independently performed the "paying the cashier" step of the task, and during three sessions he independently performed "standing in line." When video modeling was introduced during intervention he demonstrated an increasing trend with a low degree of variability and a mean performance of 3.5, with a mean change of 2.2 in performance levels between baseline and intervention phases. No overlap between baseline and intervention data was observed, indicating a PND score of 100%. During all intervention sessions, he independently performed "standing in line," "paying the cashier," and "retrieving purchased items." He did not demonstrate "smiling" during the entire intervention phase. During three intervention sessions, he independently performed "greeting the cashier" using speech-generating output and was marked as correct. He performed "thanking" with verbal prompts during intervention phase using the speech-generating device. Nick did not demonstrate independent performance when "thanking the cashier." During generalization, his social skills decreased slightly, with a mean performance of 3.0. During generalization sessions, he independently performed "standing in line," "paying the cashier," and "retrieving purchased items." However, his overall performance remained higher—with a mean change of 1.7—than baseline levels with no data overlap between the two conditions.

Social Validity

When interviewed, students indicated they liked watching the video and said it was easy to understand. They also said they felt comfortable performing tasks in the grocery store and would continue using the acquired skills in the community. The teacher satisfaction assessment results revealed that she favored the use of video modeling. She also indicated her satisfaction with the students' skill acquisition and her consideration of using video modeling to teach skills to students with autism. Each cashier also expressed positive perceptions about students' performance, referring to how students approached them by greeting them and being social during the purchasing transaction. Furthermore, all three cashiers favored the use of the speech-generating device one student used for communicating.

Discussion

The purpose of this study was to examine the effects of video modeling with verbal prompting when teaching three students with autism and intellectual disabilities to use social skills during a purchasing activity in two grocery store settings. After viewing a video depicting using social and purchasing skills to make a purchase,

each student engaged in the target behaviors. Data were recorded on the number of social and purchasing skills each student demonstrated when engaged in the purchasing activity. Results indicated that these students were able to learn specific purchasing and social skills through their exposure to a video model and to verbal prompts prior to performing skills in the grocery store; however, two students performances varied during generalization.

The increase in social along with purchasing skills for each student revealed the effectiveness of video modeling and verbal prompts for teaching these skills to participating students; it contributes to the literature base revealing positive outcomes when video instruction was used by individuals with autism and intellectual disabilities (Alcantara, 1994; Cihak et al., 2007; Haring et al., 1987; Westling et al., 1990). Although the literature on video modeling instruction included step-by-step verbal instructions in the video clip and demonstrated its effectiveness in teaching various skills to students with autism, this study examined the extent of the effectiveness of including auditory instructions in the video modeling clip. While video modeling did not include specific instructions for each task step except for waiting in line, the conversation between the adult model and the cashier was included. Individual differences among students existed. Tom was able to successfully acquire skills, while Steve and Nick did not acquire independence in the social aspects of the task.

The introduction of the intervention resulted in a rapid increase of target social and purchasing skills for Tom. During the intervention phase, Tom received verbal prompting during one session for "smiling" and one session for "greeting." He did not require verbal prompting during other sessions. He demonstrated independent performance on all steps of the task during all other sessions of intervention. The results provide insight into integrating social skills and behaviors within the context of community-based tasks for students with autism via video technology. While only Tom successfully acquired both social and functional steps of the task, future examinations with various types of modeling (e.g., adult, peer, and self-modeling) and use of video prompting delivered via portable devices, especially when interacting with unfamiliar adults in the community, are necessary. These studies might provide evidence on effective practices to support students with autism who demonstrate various levels of social and communication skills in a variety of community settings.

Increases in target social and purchasing skills between baseline and generalization conditions were observed. During generalization sessions, all students generalized purchasing skills to a second untrained grocery store. While overall increases in these target skills were observed for all students, Steve and Nick did not generalize the specific social skills they performed with verbal prompts. Both students continued to demonstrate difficulties with social skills, even after additional prompts were provided during intervention. Steve was often anxious and distracted during the task. He would pull on his clothes, flap his hands rapidly, and look around without visually attending to the task at hand. Greeting and thanking skills for Nick were inconsistent, and he often refrained from using his speech output device. Thus, examining strategies to reduce student anxiety prior to engaging in social exchanges might improve future social interactions. In addition, providing multiple opportunities for students who use alternative communication modalities to practice using them in social exchanges might increase the regularity of their use in future occasions.

Research using video modeling to teach community-related tasks implemented a 15-minute latency between watching video and performing tasks and found it to be effective with children with autism (Alcantara, 1994). Video modeling presented immediately prior to performing the task in the community versus video modeling using intervals of one hour and five minutes to one hour and 35 minutes were effective for skill acquisition for students with a moderate intellectual disability, although some of the students preferred using immediate video modeling (Taber-Doughty et al., 2008). In the present investigation, a one-hour latency between viewing the video model and performing the task was used. Thus, latency between video modeling and task performance might possibly explain Steve and Nick's performance. Future research is necessary to identify the effects of latency between instruction and task performance on students' skill acquisition. Steve and Nick might not have remembered some aspects of the video clip. Future research might examine whether watching a video clip several times or discussing the video clip in the classroom and in the community immediately prior to engaging in the task would increase skill acquisition.

The results of this study illustrate the effectiveness of video modeling and verbal prompts as an effective means for incorporating social skills when teaching autonomous behaviors in natural environments with students diagnosed

with autism and for teaching them to interact with unfamiliar people. Unlike previous studies examining video modeling, the present study combined teaching social skills along with purchasing skills in two grocery stores. The results further support previous research on the need to broaden the purchasing activity and incorporate other important skills (e.g., social greetings and responses and other socially important behaviors) to fully integrate students into the community and prepare them for independent life (Morse et al., 1996). Similar to previous investigations (e.g., Alcantara, 1994; Westling et al., 1990), this study found video modeling and verbal prompts to be an effective method of teaching new social skills along with purchasing skills in community settings to students with autism.

Limitations and Future Directions

One limitation of this study was the small number of participants. Replicating this study with a larger number of participants might produce further insights into teaching social skills and behaviors within functional tasks in community settings. Future studies might examine the impact on increasing social skills during functional activities when augmentative or alternative modes of communication are included in video models. Studies also might determine the impact of speech ability on the acquisition of social responses and skills by comparing the performance of students with and without limited speech after viewing a video model. Finally, collecting data on how people usually interact with cashiers at the grocery store might provide greater insight into the necessary skills students with autism and moderate intellectual disabilities need to be fully integrated into society.

The intervention was effective for Tom, who performed all task steps independently. However, neither Steve nor Nick performed the social skills independently or generalized them to a second untrained setting. As such, future studies might investigate if a video prompting strategy might be more effective than video modeling. Additionally, future research is necessary to identify the effectiveness of providing audio instruction in the video. For example, including specific step-by-step instructions in video modeling clips should be examined. Additionally, investigators should determine whether engaging students in a discussion of the video model prior to performing the task in the community improves skill acquisition. Investigators also might determine if providing students with more opportunities to practice skills in one setting leads to effective skill acquisition and generalization.

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